Introduction to machine learning - 4 -

Linear regression

$\underline{y} = \underline{\underline{W}}\,\underline{x} + \underline{b}$

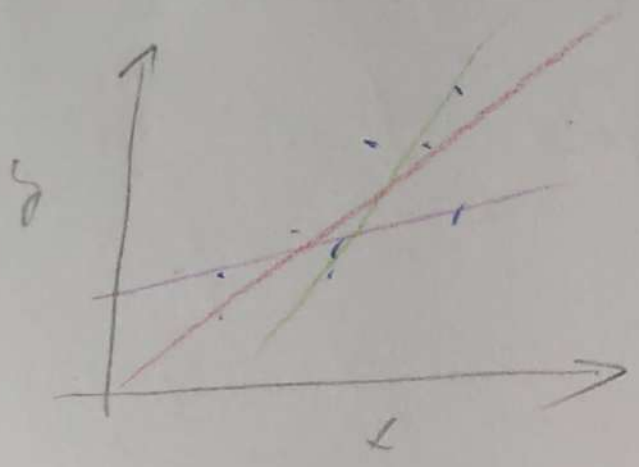
y: prediction
W: weight
b: bias

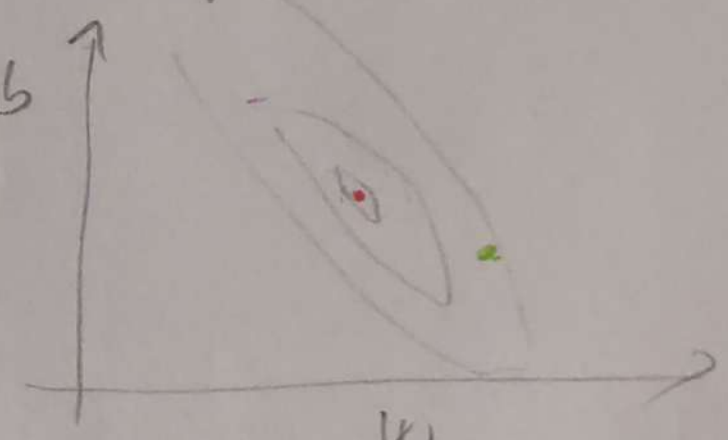
set of $\{W, b\}$ is called hypothesis, t: target

Loss function $\mathcal{L}(y, t) = \frac{1}{2}\|y - t\|^2$

Cost function $J(W, b) = \frac{1}{2N}\sum_{i=1}^{N} \|\underline{\underline{W}}\,\underline{x}^{(i)} + \underline{b} - \underline{t}^{(i)}\|^2$

N number of data points

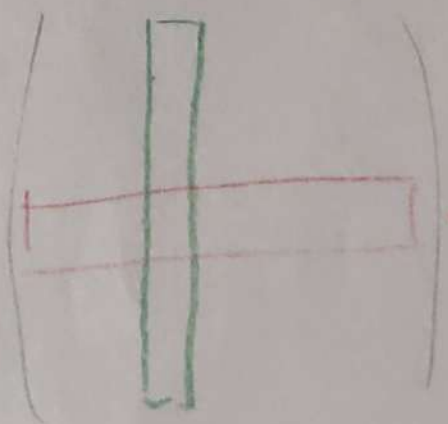




one feature (e.g. age in Titanic)

Vectorization

$$X = \begin{pmatrix} x^{(1)T} \\ x^{(2)T} \\ \vdots \\ x^{(N)T} \end{pmatrix} =$$

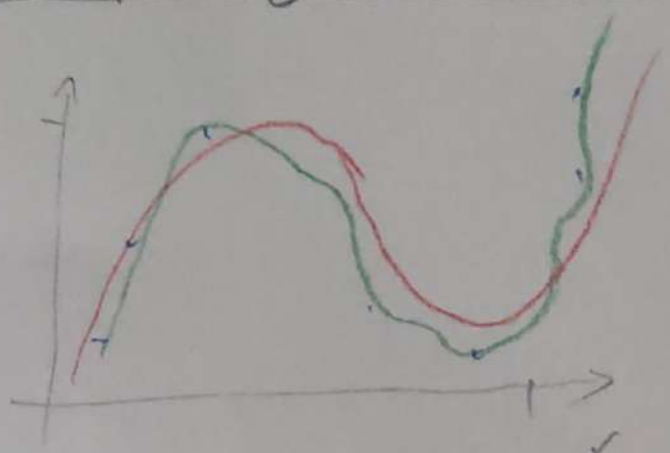


← one input vector (e.g. di Caprio)

Solution:

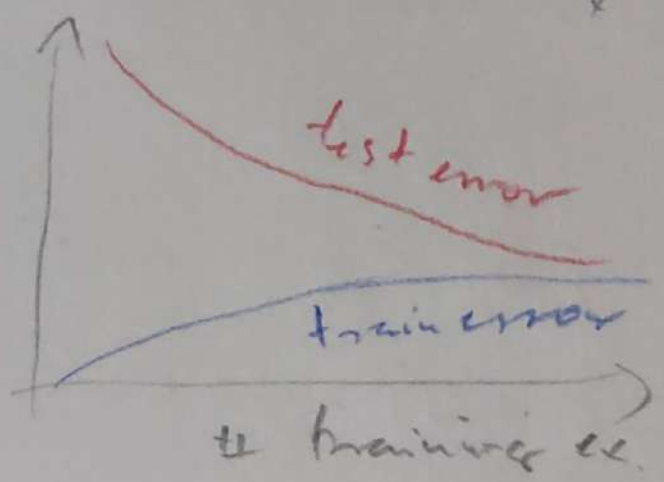
$$W = (X^T X)^{-1} X^T \cdot t$$

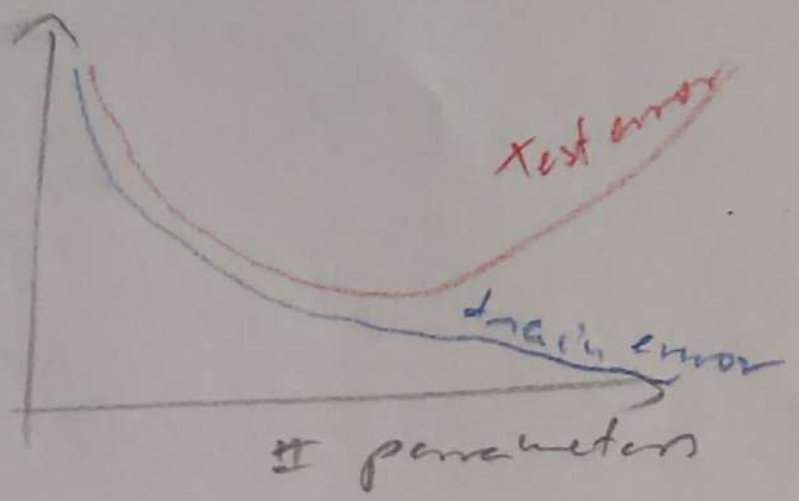
Overfitting and regularization



$\max(w) = 1$

$\max(w) = 12$





What if we need many parameters?

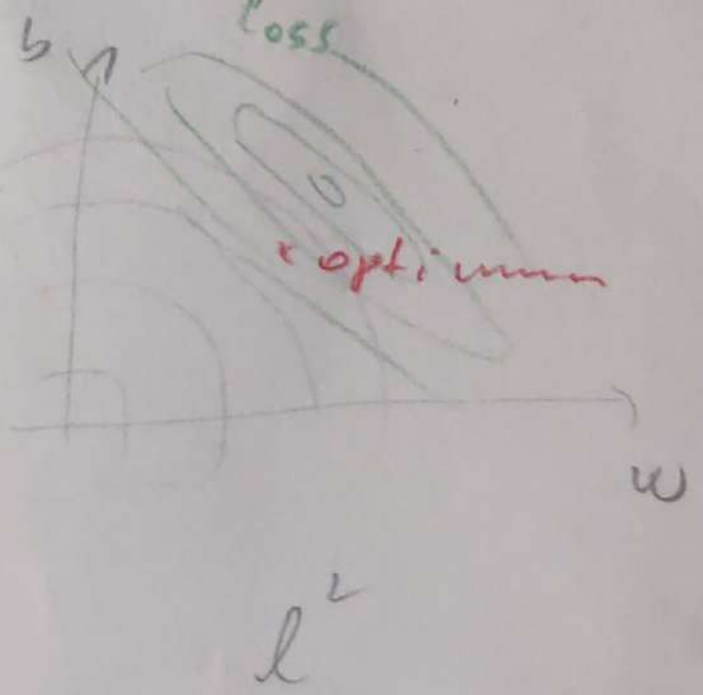
Regularization:

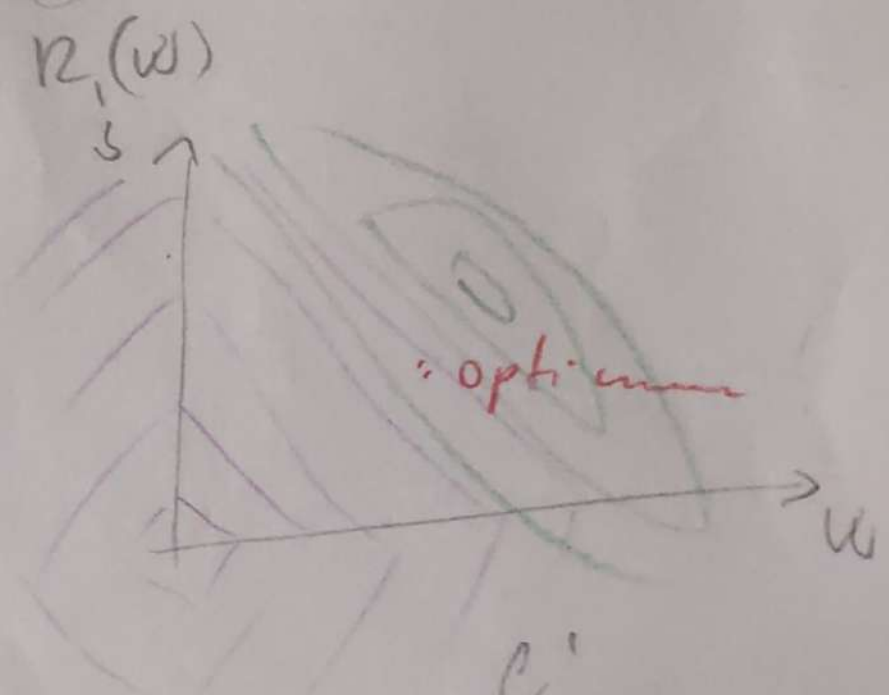
$$I_{reg} = I + \lambda \cdot \underbrace{\sum_j w_j^2}_{R_2(w)}$$

l^2 regularization

$$I_{reg} = I + \lambda \cdot \underbrace{\sum_j |w_j|}_{R_1(w)}$$

l^1 regularization





Gradient descent

$$w \leftarrow w - \alpha \frac{\partial I_{reg}}{\partial w} = w - \alpha \frac{\partial I}{\partial w} - \alpha \lambda \frac{\partial R}{\partial w} =$$

$$= (1 - \alpha\lambda) w - \alpha \frac{\partial I}{\partial w}$$

Binary data

$$z = w^T x + b \qquad y = \{0, 1\}$$

Threshold solution:

$$y = \begin{cases} 1, & \text{if } z > 0 \\ 0, & \text{if } z \leq 0 \end{cases}$$ / any value can be pushed into b)

Get rid of bias

$$\underline{x} = x_1 x_2 \dots x_n \rightarrow \underline{x} = (\underset{\uparrow}{x_0} x_1 x_2 \dots x_n)$$

Example 1:

AND

x_0	x_1	x_2	t
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

$w_0 < 0$

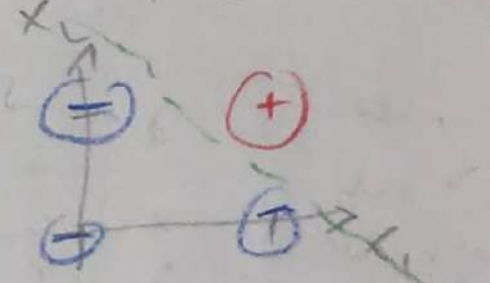
$w_0 + w_2 < 0$

$w_0 + w_1 < 0$

$w_0 + w_1 + w_2 > 0$

$w_0 \; w_1 \; w_2$

$-w_0 > w_1, w_2 \qquad -2w_0 < w_1, w_2$ ex: $-1,5; 1; 1$



Separable

Example 2 XOR

x_0	x_1	x_2	t
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

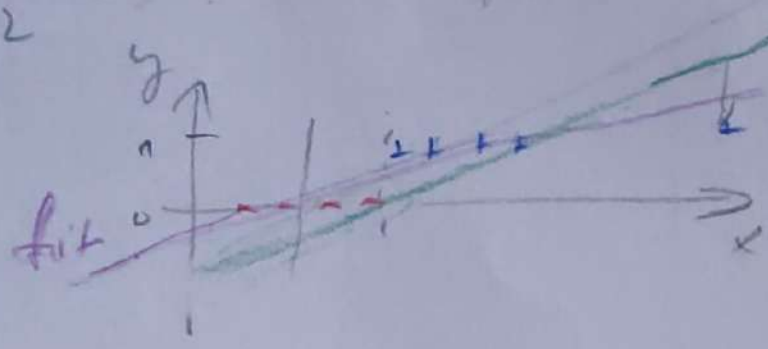
$$\left.\begin{aligned} w_0 &< 0 \\ w_0 + w_2 &> 0 \\ w_0 + w_1 &> 0 \\ w_0 + w_1 + w_2 &< 0 \end{aligned}\right\} \lightning$$

non separable

Let us solve the problem by minimizing loss!

What shall be the loss?

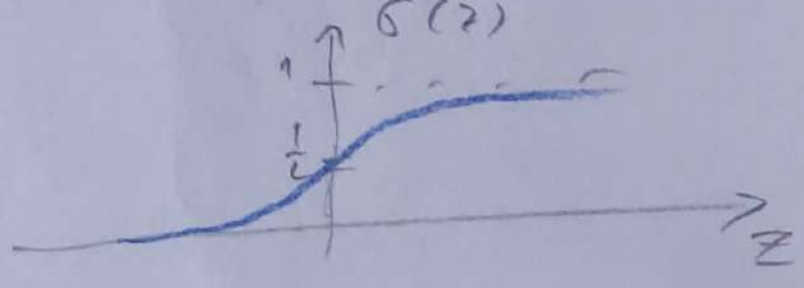
$L = \frac{1}{2}(y - t)^2$



$L = \begin{cases} 0 & \text{if } y = t \\ 1 & \text{if } y \neq t \end{cases}$ → flat is no gradient!!

Solution: Logistic activation function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

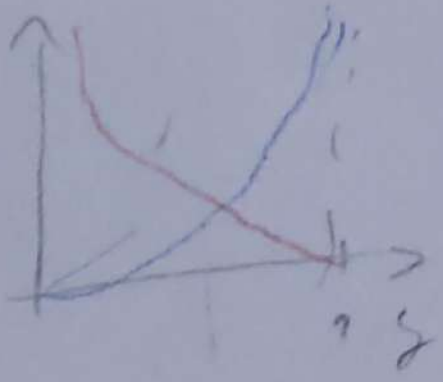


Loss function:

- $L_{SE}(y, t) = \frac{1}{2}(y - t)^2$ → problems for $z \gg 0$, $z \ll 0$ vanishing gradient

Cross entropy loss:

$$L_{CE}(y,t) = \begin{cases} -\log y & \text{if } t=1 \\ -\log(1-y) & \text{if } t=0 \end{cases}$$



$$L_{CE}(y,t) = t \cdot \log y - (1-t)\log(1-y)$$

Logistic regression

$$z = W^T x + b$$

$$y = \sigma(z)$$

$$L_{CE} = -t \log y - (1-t)\log(1-y)$$

$$W \leftarrow W - \frac{\alpha}{N}\sum_{i=1}^{N}(y^{(i)} - t^{(i)})x^{(i)}$$ exactly as in lin. reg.

Multiclass classification

one hot vectors: $t = (0\ 0\ \ldots\ 0\ 1\ 0\ \ldots\ 0)$ ← kth position for class k

$$z_k = \sum_j W_{kj} x_j + b_k$$

$$\underline{z} = \underline{\underline{W}}\,\underline{x} + \underline{b}$$

Softmax activation: $y_k = \dfrac{e^{z_k}}{\sum_{k'} e^{z_{k'}}}$

$$L_{CE}(\underline{y}, \underline{t}) = -\sum_{k=1}^{K} t_k \log y_k = -\underline{t}^T(\log \underline{y})$$

Softmax regression:

$$\underline{z} = \underline{\underline{W}}\,\underline{x} + \underline{b}$$

$$y = \mathrm{softmax}(\underline{z})$$

$$L_{ce} = -\underline{t}^{T}(\log y)$$

$$\frac{\partial L_{ce}}{\partial \underline{z}} = y - \underline{t}$$